CS3 - Case Study Rubric

DS 4002 – Spring 2026
Due: TBD
Submission format: upload your PDF and GitHub link to Canvas

Individual Assignment

General Description: Submit a link to your repo and a written PDF to the Canvas course page.

Why am I doing this? This case study is an opportunity for you to explore time-series forecasting models on a real-world crisis: climate change, and how the economy is related to it.

What am I going to do? You will explore if GDP and CO2 are linked to the contribution of global warming over time. You will clean the data sets provided, as well as pick a model that relates to time-series forecast and apply the model to these datasets. You provide deliverables that explain your reasoning for picking your choice of model as well as the code that captures the relationship between GDP and CO2.

Deliverables include:

- Repo containing the results, outputs, cleaned datasets, and source code/notebooks.
- Written work delivered in PDF format.

How will I know I have Succeeded? You will meet expectations on this Case Study when you follow the criteria in the rubric below.

Spec Category	Spec Details

Formatting	• Written Work
	  ○ Your written work should be submitted as a PDF document to Canvas under the assignment drop box.
	  ○ The written portion should be 12 pt and double spaced, along with your name, date, and instructor as the header.
	  ○ The **last** page of the written portion should contain your references. It should begin on its own page.
	  ○ You must also use in-text citations throughout your written report that references to the correct resource.
	• Repository Structure
	  ○ The repository should contain the DATA folder along with the starter README. You will have to modify the README to contain the **structure** of your repo and the **steps to reproduce** your results.
	  ○ Results should be saved in a directory titled “OUTPUTS”.
	  ○ Put all source code in a folder called “SRC”.
	  ○ Submit your GitHub link to the Canvas assignment.

Written Work	● Goal: This pdf should make the reader understand the synopsis of your plan and challenges you faced in this case study ● The written work should be no more than 3 to 4 pages, and should contain the following: ○ The first paragraph should explain the initial steps to process and clean the data as long with some EDA you have done. Explore what you find interesting about the data. ○ The second paragraph should go over the techniques you used to implement this case study. A high-level overview would suffice. ○ The next paragraph explains key results you found when working on this case study. Attach key figures and plots here. Elaborate on any trade-offs you had to make. ○ Finally, the last paragraph should go into future works, and you can use this opportunity as a brief retrospective reflection on what to do better next time.
Code	● As part of the EDA, your code should generate plots with correctly labeled axes and accurate scales. The EDA is in .ipynb notebook format. That way, the visualizations couple with your code in a self-contained file. Be sure to select at least 5 countries to visualize their GDP and CO2. ● The data cleaning scripts should output a cleaned dataset in DATA with the suffix *gdp_cleaned.csv* or *co2_cleaned.csv*. Be sure to use comments to describe your cleaning procedure if not elaborated enough in the written work. ● For each model or approach, you should create a .py script that implements it. All results can be saved in OUTPUTS. ● Your code needs to include tunable parameters for the train/test split. ● Be sure to add debug messages to notify the user which stage your model is currently in for debugging and ease-of-use.
References	● Any techniques, models, and evaluation metrics used should be cited in your written report following the format specified above. ● You need not cite the default reference materials already provided. ● Please cite it in IEEE format.

Acknowledgements: Special thanks to Jess Taggart from UVA CTE for coaching on making this rubric. This structure is pulled from Streifer & Palmer (2020).